

College of Engineering Educational Policy Committee Meeting Minutes: 2016

Minutes of the Tuesday, January 26, 2016

College of Engineering Educational Policy Committee Meeting

Time: 12- 2pm

Location 115 SEO

In Attendance: Vahe Caliskan, Houshang Darabi, John Hetling, Eduard Karpov, Joseph Hummel, Ludwig Nitsche, Alan Zdunek

1. **Meeting minutes** from December 3rd meeting **Approved** with changes.
2. **ENGR100 Revisions:** Curriculum revisions discussed at 12-3 EPC meeting were brought up at FESP committee meeting on 1-24. FESP committee agreed with these revisions:

(1) 6 weeks of general sessions with all freshman attending for the industry related seminars (Departments responsible for one speaker: BioE, ChE, CME, ECE/CS, and MIE (2 speakers)).

(2) 3 weeks of department related information (organized by the Dept DUGS)

(3) 6 weeks Department specific/FESP technical project (Organized by FESP)

Other discussion:

- A large lecture hall room has been reserved for speakers (SES 250). ENGR100 time is tentatively changed to T/R 3:30-4:45pm but time conflicts with other department courses were discussed (DUGS to check if any course conflicts at this time (verify by Wednesday 1-27))
- Transfer students can watch video during 6-week seminar sessions (since speaker will be at Freshman-only session)

3. **Math revisions:**

- a. Revision of Math 310 to eliminate Math 210 prerequisite; **EPC committee approves this change**
- b. LCN met with math department about cross-listing specific Engineering courses (IE 342, ECE 341) with math courses. Objective is to satisfy ABET math requirements (2 credits needed due to math department credit hour changes. Math department is OK with this. In order to accomplish this, there are several suggestions: 1) engineering department must change syllabus/curriculum to add more mathematical concepts to be accepted by math department and ABET; 2) Math department creates a new course on probability (2 credit hours) that becomes an introductory course for higher level statistics courses like IE342 and that satisfies 128 credit hours. Also, change Stat 381 to 2 or 3 credit hour class.
- c. **DUGS must react quickly to fix deficit in major rubric credit hours per Math revisions if not already done so, by Feb 22.** EPC to meet again on February 9 at 2pm.

4. **BioE graduate course revisions: Approved by committee**

- a. Masters program: MS candidates to be admitted must have 5 math courses (Calculus 1-3, Diff. Eq., Linear Algebra)
- b. PhD program: same as masters above, plus add core BioE courses (2 from BIOE 440, 530, 581), reduce BIOE 595 to 1 hour, eliminate qualification exam, add requirement to preliminary exam, eliminate 2X requirement to present.

5. **CME Certificate program creation: Approved by committee**

Create a certificate program as a level between UG and Masters in

- 1. Structural Engineering
 - 2. Transportation Engineering
 - 3. Water-Environmental Engineering
6. CS Graduate program revision: Table to next month
7. Information from Jim Muench: Make sure enrollment caps in department match course caps.
8. Information item: Campus Course 120 can be taken by upper-class students
9. LAS 493 internship: Information given at meeting
10. ROTC courses proposed for Gen Education categories
- a. 5 MIL courses: 2 in Understanding Individuals & Society, 2 in Understanding US Society, 1 in Understanding the Past.
 - b. **EPC approves all 5 as gen eds., no cap in COE**
11. CS109/CS107 articulation of AP credit:
- a. Possible solution for students who don't have Matlab but have AP credit for CS111; create 1-credit course that covers last 5 weeks of CS109. Joe Hummel will check to see if this is feasible.
12. ENGR space and equipment in Roosevelt building for Engineering societies/students projects is progressing:
- a. Create safety/equipment course to learn how to use equipment
 - b. Tie-in with FESP technical project course- make an option for ENGR100?

Minutes of the Tuesday, February 9th, 2016

College of Engineering Educational Policy Committee Meeting

Time: 2-4pm

Location 115 SEO

In Attendance: Vahe Caliskan, Houshang Darabi, Eduard Karpov, Joseph Hummel, Ludwig Nitsche, Alan Zdunek

1. **Meeting minutes** from January 26th meeting: **Approved** with changes.
2. **ECE Revisions: ECE 496 and ECE 497**
 - a. Discussion about how to make courses fit with Chinese 3-2 program
 - b. Wait for feedback from Derong Liu (by next week's EPC meeting)
3. **MIE Revisions: ME 320, ME 396, ME 447, IE 442; Approved by committee**
 - a. 320/442/396: update prerequisite list and list credit hours in general description
 - b. 447 --> change to 347
 - c. Academic program changes; make revisions to document before sending to LCN.
4. **CME Revisions: CME 421, CME422, CME583; Approved by committee**
 - a. **CME 583:** Add to CRS that making this a permanent course from a CME 594 course
5. **CHE Revisions: CHE 397, and Program changes due to MATH/CHEM changes; Approved by committee**
 - a. Make revisions to CRS and academic program change document before sending to LCN
 - a. Revisions to program to included MATH and CHEM changes
6. **ECE suggestion to fulfill Math requirements:** Vahe will discuss with ECE UG committee and send out for approval to EPC by email vote
7. **CS Revisions and Program changes:**
 - a. **PhD program revisions (direct admits and those with MS) Approved by committee**
 - b. It is feasible to take 1-credit hour course that covers Matlab (as a part of CS109) for those who have CS111 AP credit or don't have Matlab.
8. **MakerSpace/FabLab: Student usage and safety courses**
 - a. Discussion: Committee suggested creating a general safety course that everyone using the facility must take. The course should cover safety information regarding the facility and regarding all equipment. There should also be specific courses created for the three equipment areas; 1) additive (3D printing, CNC), 2) subtractive (milling etc), 3) electronic so that the students can learn how to use the equipment properly.
 - b. All engineering students can potentially use the facility as long as they register in the COE office (record-keeping) and take the safety and usage courses above.
 - c. Discuss further at next week's meeting (CRS's for courses)
9. **International Engineering minor:**
 - a. EPC comment: Don't focus on one particular language/culture, make it more general

Minutes of the Tuesday, February 16th, 2016

College of Engineering Educational Policy Committee Meeting

Time: 2-4pm

Location 115 SEO

In Attendance: Vahe Caliskan, Houshang Darabi, Eduard Karpov, Joseph Hummel, Ludwig Nitsche, Alan Zdunek

1. **Meeting minutes** from February 16th meeting: **Approved** with changes.
2. **ECE revisions due to MATH 180/181 changes: CE and EE programs; Approved by committee**
 - a. Students will make up 2 credit hours with another math course from the list: MATH 310, 410, 417, 481, MCS 471 (2 required from this list)
 - b. Updated programs with above change plus chemistry course number change and Elective Outside the Major Rubric credit hour change from 3 to 2 hours.
 - c. CS 385 in Technical Electives; Vahe to get dept approval about removing and adding CS 361
3. **CHE revisions to Minor in ChE and Honors program; Approved by committee**
 - a. Check with Chemistry dept whether students can take CHEM 222 after taking CHEM 116/118.
4. **CS revisions to BS program; credit hour reduction: Approved by committee pending items below.**
 - a. Revision: Reduce total hours to 120; Decrease due to Math changes plus reduce Free Electives to 6 hours
 - b. EPC approves from an educational perspective pending a check of whether the bylaws have any procedures about changes in program hour requirements. EPC suggests the changes to be considered by Administration Council for final approval.
 - c. Two other issues; 1) decreasing hours can be perceived as weakening the program, 2) less hours means less revenue to university.
 - d. Approval by EPC pending
5. **CS revision to CS 109: Approved by committee**
 - a. Revise credit hours from 3 to 1 to 3 (C/C++ is 2 credit hours, MatLab is 1 credit hour), both together are 3 credit hours.
6. **MakerSpace/FabLab Courses:**
 - a. **ENGR112 Approved pending question to be answered below:**
 - b. Make the grading S/U or normal grading (A-F); ask instructor for feedback; EPC to approve by email
 - c. ENGR113: Broaden title to allow for new topics to be considered later on.
7. **MIE Program due to MATH changes: Pending**

Minutes of the Tuesday, September 6th, 2016

College of Engineering Educational Policy Committee Meeting

Time: 1-3pm

Location 115 SEO

In Attendance: Houshang Darabi, Natasha Devroye, John Hetling, Joseph Hummel, Ludwig Nitsche, Alan Zdunek

1. **Election of Chair and Secretary:** Joe Hummel was elected as chair for 2016-2017, Al Zdunek elected as secretary for 2016-17, both for second terms.
2. **EPC meetings:** Motion to make a regular meeting time: Every other Tuesday at 1pm, unless cancelled. Motion carries.
3. **MIE Program changes have been approved:**
 - a. Change to EM program: IE473 added, IE441 removed, sample schedule revised
 - b. Change to IE program: List of IE Tech electives modified: ME447
 - c. ME change: Add CME434 to technical electives: CME dept will be notified
4. **Discussion on ME444/ME445** equivalent to ME 396/ME397, tabled into next meeting
5. **ME308 pre-requisite change:** Include CS109 and drop other pre-reqs except possibly keep MATH220 (HD will double-check to make sure MATH 220 pre-reqs are not needed)
6. **ME312 change:** Add CS109 and ECE210 and remove MATH220 and PHYS142 (these are implicit due to ECE210)
7. **ME428 change:** Remove CS108 and add CS109
8. **GPA Recalculation Policy policy change:** EPC agrees with the policy change, with a change to one paragraph on page 5 to make it clearer that it is the lower grade of a repeated course that will be excluded from the GPA. Jim Muench will make the revision and EPC will vote by email to approve.
9. Discussed/brainstormed Manufacturing Connect Partnership /student mentors program; like the UIC/Austin HS mentoring program, expanded to all UIC students, more HS's, make it a 1-credit hour course. Further discussion to be continued.
10. New Business:
 - a. Pre-requisite enforcement brought up by John Hetling, college-wide form to approve pre-requisites

Minutes of the Tuesday, October 18th, 2016

College of Engineering Educational Policy Committee Meeting

Time: 1-2:30 pm Location 115 SEO

In Attendance: Houshang Darabi, Natasha Devroye, John Hetling, Joseph Hummel, Eduard Karpov, Ludwig Nitsche, Alan Zdunek

1. **Minutes from September 20th meeting approved** with correction to Grade Recalculation discussion
2. **BioE items:**
 - a. Course name change: BIOE/BPS 518 to Advanced Drug Delivery Systems: Discussion regarding credit hours: Proposal shows 3 for grad students, 2 for professional students, but normally Engineering grad courses are 4 credit hours. **EPC approves BIOE 518** but with a change from 3 to 4 credit hours for graduate students, 2 for professional students. BPS 518 stays as 3/2 credit hours. John Hetling will ask instructor about revising Course Major Topics to better reflect the current course topics.
 - b. Revision of PhD program in BioE: Previously approved by EPC but there was a change in the revision afterwards; 1) no publication before prelim and 2) Particular courses required previously were taken out. **EPC agrees to table topic** until John discusses with David Eddington.
3. **CHE 451 new course and Change to BS CHE program** to add CHE 451
 - a. **EPC approves CHE 451** with changes to Learning Outcomes, Learning Outcome Assessment Methods, Major Topics
 - b. **EPC Approves** Revision to Chemical Engineering program to add CHE 451 as a technical elective
4. CS 110 and Matlab prerequisite enforcement tabled until next meeting
5. **(X94) ENGR general engineering courses**
 - a. 294 and 394 technical elective courses: These courses are created to have selected topics for sophomores and juniors like the 494 senior-level courses. First area is for ENGR makerspace use, but can also expand out to departments. **EPC approves**
6. New business:
 - a. Online advising system
 - b. Evaluation of the faculty evaluation system
 - i. Discussion, evaluate internal system and research other college evaluation systems; will need an internal champion, maybe the TLC (scholars).

Minutes of the Tuesday, November 1, 2016

College of Engineering Educational Policy Committee Meeting

Time: 1-3pm

Location: 115 SEO

In Attendance: Houshang Darabi, Natasha Devroye, John Hetling, Joseph Hummel, Eduard Karpov, Ludwig Nitsche, Alan Zdunek

1. Minutes from October 18 meeting approved, Sept 20th meeting minutes revised (Bridge Design 'I' added)

2. China 3+2 item (CME and ECE):

- a. China 3+2 program discussion presented by Prof Goncharoff on creating a more formal course system in program. Right now Departments with active program are using two 49X temporary courses (8 hours each semester) that satisfy China requirements.
- b. **Discussion:** Departments to create two courses (fall/spring) called UG Senior Design Thesis I and II. Course I emphasizes the design part and course II emphasizes the thesis portion as per China's suggestion. Further suggestions on how to create the CRS for each course are summarized below from Eduard Karpov's email on November 2, 2016 to Derong Liu:
 - i. Preferred course titles placing them in line with other thesis-type courses, such as 598 and 599, should be (please see if these will work for the program):
 1. Undergraduate Senior Design Thesis I and
 2. Undergraduate Senior Design Thesis II
 - ii. Number of credits for both courses should be variable: 0-8
 - iii. CRS course description of the two courses (I & II) can be identical but it should be a merge of the current two course descriptions AND the current 598/599 (MS/PhD thesis) course descriptions.
 - iv. The term "Certificate" should be avoided in the CRS.
 - v. The term "Certificate" should be clarified to the committee for the purpose of discussion. The committee was concerned that it cannot be a UIC issued certificate.
 - vi. Section "General reason for this request" should contain more information for the SCEP members beyond CoE of the type "To facilitate international programs of the College ..., to address broader international audience..." and similar.
 - vii. Section "Purpose of the course in relation..." should not say that the courses is similar to Senior Design.
 - viii. Section "Major Topics" should probably be blank.
- c. This item is tabled until Eduard gets feedback from Derong and ECE questions are answered; it will be brought up at the Nov 15th meeting.

- i. Derong will be invited to the next EPC meeting (Nov 15); new draft of CME 498/499 and drafts for all dept CRS's

3. CME items:

- a. **CME 434:** change to crosslist with ME course, and change pre-requisites ; **EPC approves with addition of learning outcomes**

4. BioE items:

- a. **New course BIOE485 Nanobiosensors:** discussion about crosslisting with ECE; question about whether prerequisites can be different for crosslisted courses. Ludwig will check on pre-reqs and Natasha on ECE crosslist. **EPC approves**

5. ECE items:

- a. Addition of graduate college restriction' to all ECE 500-level courses, **EPC approves**
- b. **ECE 535;** course taught by MENG and is only for MENG students, add "course restriction to MENG students"; **EPC approves, with revision of small format errors.**

6. Old business:

- a. **BIOE graduate requirements:** John Hetling discussed with David Eddington the pushback from SCEP on several points; David agrees to remove the items, EPC previously approved document and agrees that BIOE can move forward, latest proposal to be sent to Ludwig.
- b. **CS 110:** 1-hour Matlab course for transfer students who have programming course already. Questions were asked by Monica Soto and answered, course approval should move forward. Discussion that if transfer students have programming only they get credit for CS111 and must take CS110 (matlab). Once approved, each department should change their CS requirement to include CS110/111 or CS109.

7. New business:

- a. **Curriculum credit hour requirements:** Credit hour decrease for ECE to 124; CS dept is thinking about 126 hours. EPC cannot make changes, must be approved at full college faculty meeting.

Minutes of the Tuesday, November 29, 2016

College of Engineering Educational Policy Committee Meeting

Time: 1-3pm Location 115 SEO

In Attendance: Houshang Darabi, Natasha Devroye, Joseph Hummel, Eduard Karpov, Ludwig Nitsche, Alan Zdunek, Angus Forbes (guest)

6. Minutes from November 1 meeting approved, change slight format error.

7. Changes to Undergraduate catalog (Ludwig Nitsche):

- a. Differences in Engineering majors taking a minor in Math; some take more or less courses, CS has no pathway.
- b. Proposal to open a pathway for CS majors to get a math minor; by taking two extra Math courses: **EPC approved.**
- c. There are other issues on course discrepancies that will be discussed at a later EPC meeting.
- d. Discrepancy in double majors: Undergraduate Catalog formally states that Engineering majors cannot receive a major from another college; however this should be allowed. Further discussion needed to propose this change.

8. CS Items:

- a. An MS/MFA joint program in CS and New Media Arts (CSA) was presented by Angus Forbes. Discussion and good agreement on creating the joint program
- b. **EPC approved with minor changes to be made on CRS form.**

9. MIE items:

- a. **IE 473:** Change pre-req from Math220 to Math 210 to allow Engineering Management majors to take course and without having to take MATH220. This course is required for Eng Mgmt majors, but they don't need MATH220. **EPC Approved**
- b. **ME525:** Add teaching content in CRS to better reflect course **EPC approved**

10. China 3+2 program courses

- a. ECE and ChE have course descriptions but they have not been approved yet by Derong Liu. EPC decided to use the CHE course as a template; EPC to review it and use as a template for other engineering department 3+2 course CRS's.
- b. **Sharepoint:** Ludwig suggested using Sharepoint to store EPC documents; EPC must obtain access, Ludwig will put CHE 496/497 in EPC Sharepoint folder for everyone to edit.

11. Old business: None

12. New business: Next meeting Tuesday December 13, 2016 with Derong Liu

Minutes of the Tuesday, December 13th, 2016

College of Engineering Educational Policy Committee Meeting

Time: 1-3pm

Location 115 SEO

In Attendance: Houshang Darabi, Natasha Devroye, Joseph Hummel, Ludwig Nitsche, Alan Zdunek, Derong Liu (guest), Daniela Tuninetti (guest), (other guest)

- 1. Minutes from November 29th meeting EPC approved**
2. Sharepoint EPC folder up and running with documents organized by meetings and departments
- 3. China 3+2 program courses (496/497)**
 - a. Derong Liu discussed how the CRS documents should be put together for international programs like the 3 + 2 China program. Discussion focused on ECE documents; changes were discussed for several of the sections. **EPC approved ECE 496/497 with these changes as discussed;** LCN to revise ECE 496/497 and upload to Sharepoint for use as a template for other departments.
- 4. ECE Curriculum Revisions:**
 - a. 7, BS in Engineering Physics Degree requirement changes as outlined in proposal document were discussed. Degree requirements will change significantly and require revision of the ABET-related student outcomes. There was a minor change to Change #7 wording. It was also suggested that engineering departments affected by these changes be notified. **EPC approves.**
 - b. 10 BS degree changes to CE and EE: **EPC approves with revisions as discussed.**
 - i. Change 1 removes ME 205/CHE 201. Suggestion was made to inform MIE and CHE depts about this change
 - ii. Changes 2-6 were discussed without any major revisions/issues; suggestion was made to just mention other Math courses
 - iii. Change 7: Make courses mandatory with other courses moved to technical electives.
 - iv. Change 8: ECE students have shown interest in ENGR 400 courses, but they are not technical electives so they can't be in this category. Also they are online courses; COE policy on online courses is not certain. This change will be removed.
 - v. Change 9, OK,
 - vi. Change 10; use same wording as in Engineering Physics revision.
 - vii. Revisions to Notes at end.
- 5. Math minors update;** ongoing.
- 6. ENGR community engagement courses:** work is proceeding to create ENGR 497/498 courses by Fall 2017